

DTSA 5842

# Effective Communication: Writing, Design, and Presentation

|                                        |                                        |               |
|----------------------------------------|----------------------------------------|---------------|
| FORMAT                                 | CATEGORY                               | CREDITS       |
| Online (DTSA)                          | Elective                               | 1             |
| SECTION                                | BEST ROLE MATCH                        | AVG ALIGNMENT |
| Section 4 – Elective                   | Machine Learning Engineer (54.3 / 100) | 47.1 / 100    |
| SPECIALIZATION Effective Communication |                                        |               |

47

CURRENT AVG SCORE  
across 6 job roles

83

POTENTIAL SCORE  
with recommended additions

+55

MAX SCORE GAIN  
best single-role improvement

CURRENT COURSE DESCRIPTION

This course teaches students how to present themselves effectively through writing, design, and presentation. Students will focus on how to write well-organized, clear business documents; to design elegant presentation slides, reports, and posters; and to present and speak with confidence and power. More broadly, the course charts a journey toward each student’s best professional self. This course is a prerequisite for the Effective Communication Capstone.

INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE |      | PROJECTED SCORE |       | POINT GAIN | PRIORITY |
|---------------------------|---------------|------|-----------------|-------|------------|----------|
| Machine Learning Engineer | <div></div>   | 54.3 | <div></div>     | 84.7  | +30.4      | HIGH     |
| Applied Scientist         | <div></div>   | 50.2 | <div></div>     | 100.0 | +49.8      | HIGH     |
| ML Engineer               | <div></div>   | 49.5 | <div></div>     | 73.6  | +24.1      | MEDIUM   |
| AI Engineer               | <div></div>   | 47.6 | <div></div>     | 78.9  | +31.4      | HIGH     |
| Data Scientist            | <div></div>   | 45.3 | <div></div>     | 100.0 | +54.7      | HIGH     |
| MLOps Engineer            | <div></div>   | 36.0 | <div></div>     | 61.7  | +25.6      | MEDIUM   |

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

**System Design**

2 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

MEDIUM

**Consider System Design content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: systems.

GAP KEYWORDS BY ROLE

|                                                                                  |                          |
|----------------------------------------------------------------------------------|--------------------------|
| <b>Machine Learning Engineer</b>                                                 | 54.3 → 84.7 (+30.4 pts)  |
| data sets learning analytics tools exploratory mobile development                |                          |
| <b>Applied Scientist</b>                                                         | 50.2 → 100.0 (+49.8 pts) |
| discovery scientific research interdisciplinary models technical department data |                          |
| <b>ML Engineer</b>                                                               | 49.5 → 73.6 (+24.1 pts)  |
| discovery data learning machine systems models model world                       |                          |
| <b>AI Engineer</b>                                                               | 47.6 → 78.9 (+31.4 pts)  |
| perception following data models learning build software technical               |                          |
| <b>Data Scientist</b>                                                            | 45.3 → 100.0 (+54.7 pts) |
| data sets product analytics social build                                         |                          |
| <b>MLOps Engineer</b>                                                            | 36.0 → 61.7 (+25.6 pts)  |
| data model systems learning machine tools models development                     |                          |

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.